

Chemistry of transition elements

A Level Chemistry

Transition elements

A **transition element** 过渡元素 is a d-block element that forms one or more stable ions with **incomplete d orbitals**. (Scandium and zinc are in the d-block but are not transition elements, because their stable ions have empty or full d orbitals.)

The 3d **orbitals** 轨道 have set shapes: the $3d_{xy}$ orbital has four lobes pointing between the axes, and the $3d_{z^2}$ orbital has two lobes along the z-axis with a ring around the middle.

Four key properties (and why)

Property	Reason
variable oxidation state 氧化态	the 3d and 4s sub-shells are close in energy, so similar small amounts of energy remove different numbers of electrons
act as a catalyst 催化剂	they have more than one stable oxidation state, and vacant d orbitals that can form dative bonds
form complex ions	vacant d orbitals can accept lone pairs
form coloured compounds	electrons move between split d orbitals (see below)

Ligands and complexes

A transition metal ion can be surrounded by **complex ions** 配离子. The species attached are ligands.

A **ligand** 配体 is a species with a lone pair of electrons that forms a **dative covalent bond** 配位键 to the central metal ion. (The **lone pair** 孤对电子 is what it donates.) Ligands are grouped by how many such bonds they can form:

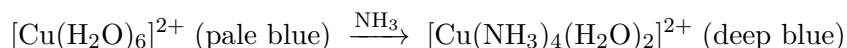
- **monodentate** 单齿: one bond (H_2O , NH_3 , Cl^- , CN^-).
- **bidentate** 双齿: two bonds (1,2-diaminoethane "en", and the ethanedioate ion $\text{C}_2\text{O}_4^{2-}$).
- **polydentate** 多齿: many bonds (EDTA^{4-} , which uses six).

A **complex** 配合物 is a central metal atom or ion surrounded by one or more ligands. Its shape can be **linear** 直线形, **square planar** 平面正方形, **tetrahedral** 四面体形 or **octahedral** 八面体形.

The **coordination number** 配位数 is the number of dative bonds from the ligands to the central ion ($6 \rightarrow$ octahedral, $4 \rightarrow$ tetrahedral or square planar, $2 \rightarrow$ linear). To predict the charge of a complex, add the metal's charge and all the ligand charges.

Ligand exchange

In **ligand exchange** 配体交换 one ligand replaces another, often with a colour change. For copper(II):



With concentrated HCl it becomes yellow $[\text{CuCl}_4]^{2-}$; cobalt(II) behaves in a similar way.

Redox reactions of transition ions

Use $E^\ominus$ values to predict whether a redox reaction is feasible. Common titrations you should be able to calculate include $\text{MnO}_4^-/\text{C}_2\text{O}_4^{2-}$ and $\text{MnO}_4^-/\text{Fe}^{2+}$ in acid (purple to colourless), and $\text{Cu}^{2+}/\text{I}^-$ (which makes iodine, then titrated with thiosulfate).

Why complexes are coloured

In a free ion the five d orbitals are **degenerate** 简并—they have the same energy. When ligands come close, they split the d orbitals into two **non-degenerate** 非简并 sets, separated by an energy gap ΔE :

- **octahedral**: three lower and two higher orbitals.
- **tetrahedral**: two lower and three higher orbitals.

A complex absorbs light whose frequency matches ΔE , promoting an electron from a lower to a higher d orbital. The colour you **see** is the **complementary colour** 互补色 of the light absorbed. Different ligands give a different ΔE , so they change the frequency absorbed and hence the colour—which is why ligand exchange changes the colour.

Stereoisomerism in complexes

- **geometrical isomerism** 几何异构 (**cis** 顺式 / **trans** 反式) appears in square planar complexes such as $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$, and in octahedral complexes such as $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$.
- **optical isomerism** 旋光异构 appears in octahedral complexes with bidentate ligands, such as $[\text{Ni}(\text{en})_3]^{2+}$, which has two non-superimposable mirror images.

You can also deduce the polarity of a complex: a *cis* form may be polar, while the matching *trans* form is often non-polar because its dipoles cancel.

Stability constants

The **stability constant** 稳定常数 (K_{stab}) is the equilibrium constant for forming a complex ion from the metal ion and its ligands in solution (water is left out of the expression).

A **large** K_{stab} means a very stable complex. In a ligand exchange, the position moves towards the complex with the larger K_{stab} —that is why a ligand that forms a more stable complex can push out a weaker one.